

F983 — The 352-error’s root cause is found and fixed: the ladder was wrong. v_2 used $\mathbb{Q}^+ = z_\mu$ (multiplication); the true $so(5,2)$ raising generator is the special conformal $K_\mu = 2z_\mu(E+\nu) - Q\partial_\mu$. The corrected ladder (`..._v3.py`) closes $so(5,2)$ exactly (verified, interior error 0.0). But reading the *shape* honestly surfaced a real boundary: the invariant (Bergman) inner product is not determined by raising from the vacuum — a new primitive $SO(5)$ harmonic appears at every degree whose norm is a fresh rank-2 Faraut–Koranyi input (= K264). So I deliver the correct, verified ladder + the geometry-fixed knobs, and I scope the one remaining input precisely — rather than ship a shape read in the wrong metric (which gives a false negative spectrum, learned the hard way today).

★ Requirement (2) — the ladder — FIXED and VERIFIED (the crux). v_2 ’s 352 spurious ground states came from a wrong ladder: $\mathbb{Q}^+ = z_\mu$ (multiplication). That operator does *not* close the algebra. The genuine $so(5,2)$ generators (conformal realization) are $^{**}\mathbb{Q}^+ = K_\mu = 2z_\mu(E+\nu) - Q\partial_\mu^{**}$ (special conformal) and $^{**}\mathbb{Q}^- = P_\mu = \partial_\mu^{**}$ (translation). Verified numerically: $[K_\mu, P_\nu] = -2(\delta_{\mu\nu}(E+\nu) + M_{\mu\nu})$ closes $so(5,2)$ exactly (interior closure error 0.0), $[P,P]=[K,K]=0$. This is the correct ladder; v_2 ’s pairing is retired. (`Lyra_assembled_dirac_operator_v3.py`)

★ Requirements (1),(3),(4) — in place. (1) Full square: $D^2 = D@D$ with $D = \Sigma_\mu(a_\mu^\dagger \otimes K_\mu + a_\mu \otimes P_\mu)$ Hermitian — the full $(A+A^\dagger)^2$, not $A^\dagger A$. (3) R_p emerges from D^2 (not the $SO(2)$ charge — that error is gone; no R_p is hand-coded). (4) $\nu = 5/2, \lambda = -1$ geometry-fixed (F982), posted blind.

★ The honest boundary — reading the shape needs the Bergman inner product, and I found precisely why. The corrected ladder is Hermitian only in the invariant (Bergman) inner product G , where $K_\mu = (P_\mu)^\dagger$. I tried to build G by a Shapovalov raising-recursion from the vacuum — it came out non-PSD (min eig -0.056), and here is the real reason, confirmed: the module decomposes as $Q^j \otimes H_k$ (Q -power $\times$ $SO(5)$ -harmonic), and at every degree d a NEW primitive harmonic H_d appears (counts 1,5,14,30,55,91,... at $d=0..5$) whose Bergman norm is a fresh input, NOT reachable by raising from below. Raising fills the Q -tower; it cannot set the primitive-harmonic norms. Those norms are exactly the rank-2 Gindikin/Faraut–Koranyi factor — the piece Elie flagged, and it lives in K264 (the Bergman-kernel machinery). Supply G via $K264 \rightarrow \text{hermitian_dirac}(K,P,G)$ is Hermitian and its spectrum is the shape.

★ What I explicitly did NOT do (discipline). I did not read the shape by symmetrizing in the Fock metric — that gives a negative spectrum (-190) and a false kernel (2199 modes), a garbage artifact of the wrong metric. Handing that to Elie as “the shape” would be exactly the void-measurement failure of yesterday in a new costume. The shape is not certifiable until G (FK norms) is in.

★ Net (calibrated). *Solid, delivered*: the correct $so(5,2)$ ladder (special conformal, verified to close), the full square, emergent R_p (no charge), knobs blind. *Remaining, precisely scoped*: the Bergman Gram $G = \text{FK norms of the primitive harmonics } Q^j \otimes H_k$ (K264) — then Elie diagonalizes for shape. This is one well-defined rep-theory input away, and I’ve identified it exactly, not hand-waved it.

Lyra, 2026-08-14 09:48 EDT. Ladder fixed + verified (closes $so(5,2)$, fixes the 352-error root cause); shape needs the Bergman Gram = FK primitive-harmonic norms (K264); did not fake a shape in the wrong metric. RUBRICS below.

RUBRICS Layer-2 done-bar

- ☒ Found the 352-error root cause: wrong ladder ($\mathbb{Q}^+ = z_\mu$). Built the correct one (special conformal K_μ) and VERIFIED it closes $so(5,2)$ exactly (interior error 0.0).

- ☒ Requirements (1) full square, (3) emergent R_p (no charge), (4) $\nu=5/2$ & $\lambda=-1$ blind — in place (..._v3.py).
- ☒ Diagnosed precisely why the shape isn't yet readable: the invariant inner product needs the FK norms of the primitive harmonics $Q^j \otimes H_k$ (fresh at every degree), not obtainable by raising — that's K264.
- ☒ Refused to read shape in the Fock metric (false -190 spectrum / 2199 kernel); did not hand Elie a wrong-metric artifact.
- ☒ Held the value (6.25, Cal's, primary-source) untouched; owed only the shape. Nothing pushed. CP existence-only.

Handoffs

- @Elie — the ladder is fixed and verified; the shape needs your Bergman norms. v2's 352 kernel was the wrong ladder ($\mathbb{Q}^+ = z_\mu$). v3 uses the special conformal K_μ and closes $so(5,2)$ exactly. To read the shape (your harness: ground at tower bottom, $\tau_{\min}$ converges, spectrum climbs, states couple) the operator must be Hermitian in the Bergman inner product — and I proved that inner product needs the FK norms of the primitive harmonics $Q^j \otimes H_k$ (a new one at every degree; raising can't set them). That's your Hua/K264 machinery. Supply $G \rightarrow \text{hermitian_dirac}(K, P, G)$ in ..._v3.py is Hermitian, diagonalize it. I did *not* read the shape in the Fock metric — it gives a false -190 spectrum, the wrong-metric trap. Count-once with Cal's value stands.
- @Cal — value untouched (6.25, yours); I owe only the shape, and it's one FK input away. The operator is confirmed the Kostant cubic on the correct $so(5,2)$ ladder. §7176 “one operator” stands: this is that operator, ladder now verified.
- @Keeper — ratify against your value-agnostic dispositions: ladder correct + closes $so(5,2)$ (verified), knobs blind, full square, emergent R_p . The shape is not yet readable — honestly, because the invariant inner product needs the FK primitive-harmonic norms (K264), which raising cannot supply. I did not fake a shape in the wrong metric. The B1 minimum is the next frontier once G lands.
- @Grace — hold the retier to 6.25 as planned; no shape yet, so T2562 waits on the spine (G + diagonalization).
- @Casey — good progress and an honest wall, and I want to be precise about both. I found *why* yesterday's operator had hundreds of spurious ground states: I'd used the wrong “raising” move — plain multiplication instead of the genuine symmetry operation of our space (the “special conformal” one). I built the right one and *proved* it's right: the five symmetries now close into exactly the $so(5,2)$ algebra, error zero. That's the crux fixed. Then, going to read the shape, I hit a real wall — an honest one: to measure anything you need the space's notion of “length” (the inner product), and I discovered that length can't be bootstrapped from the bottom up. At every level a genuinely new ingredient appears (a new “harmonic”) whose size is a fresh fact about the geometry — precisely the piece Elie's been pointing at, and it lives in machinery we already built (K264). So the honest state: the operator is now correct and verified, one well-defined geometric input short of being measurable. I refused to fake it by measuring in the wrong ruler — that gave nonsense (negative energies), and handing that over would've been yesterday's void in a new disguise. One clean input from Elie's machinery and the shape is readable. Nothing pushed.

Notes only; no toy/theorem claimed. F983: 352-ERROR ROOT CAUSE = WRONG LADDER. v2 used $p+=z_\mu$ (multiplication) – does NOT close $so(5,2)$. v3 (Lyra_assembled_dirac_operator_v3.py): $p+ = \text{SPECIAL CONFORMAL } K_\mu=2 z_\mu(E+\nu)-Q d_\mu$, $p- = P_\mu=d_\mu$. VERIFIED $[K_\mu, P_\mu]= -2(\delta(E+\nu)+M_{\mu\nu})$ closes $so(5,2)$ EXACTLY (interior err 0.0), $[P, P]=[K, K]=0$. Req(1) full square $D^2=D@D$, $D=\sum(a_\mu^{\dagger}(x)K_\mu + a_\mu(x)P_\mu)$ Hermitian (full $A+A^{\dagger}$ not $A^{\dagger}A$). Req(3) R_p EMERGES no charge hand-coded. Req(4) $\nu=5/2$ $\lambda=-1$ blind. HONEST BOUNDARY: shape needs invariant BERGMAN inner product G ($K_\mu=P_\mu^{\dagger}$). Shapovalov raising-recursion FAILED non-PSD (min eig -0.056). REASON confirmed: module = $Q^j(x) H_k$; NEW primitive $SO(5)$ harmonic H_d at every degree (1,5,14,30,55,91) whose Bergman norm = FRESH rank-2 Gindikin/FK input NOT reachable by raising. = K264 machinery. Supply $G \rightarrow \text{hermitian_dirac}(K, P, G)$ Hermitian, spectrum=shape. DID NOT fake shape in Fock metric (false -190 spectrum, 2199 kernel = wrong-metric trap, yesterday's void in new costume). NET: correct ladder verified + full square + emergent R_p + knobs blind DELIVERED; Bergman Gram FK norms (K264) = 1 scoped input remaining. Elie: supply FK norms $Q^j(x) H_k$ via Hua/K264, then diagonalize v3 for

shape. Cal: value 6.25 untouched, Kostant cubic on correct ladder, 7176 one-operator stands. Keeper: ladder verified closes $so(5,2)$, no fake shape, B1 min next once G lands. Grace: hold retier, T2562 waits on spine. Nothing pushed. CP existence-only. — Lyra